PROVAL co.,Ltd

Calculation Sheet

Calculation header

Identifier	*New calculation (1)*

Medium selection and state

Medium	*Water*
State	*Liquid*

Operating data

☐ Safety-related application

	Maximum flow	Mean flow	Minimum flow	
t1	*60.0*	*0.0*	*0.0*	°C
p1	*10.0*	*0.0*	*0.0*	MPa(g)
○ p2	*4.0*			MPa(g)
◉ Δp	*6.0*	*0.0*	*0.0*	MPa
Cv	*74.159*			GPM(US)
○ qm	*493,760.0*			kg/h
◉ qv	*500.0*	*0.0*	*0.0*	m^3/h
LpAe	*109.0*			dB(A)
	Incipient cavitation			

Fluid operating data

	Maximum flow	Mean flow	Minimum flow	
ϱ1	*987.52*	*0.0*	*0.0*	kg/m^3
ϱ2	*987.52*			kg/m^3
○ η1	*0.46907*			mPa s
◉ v1	*0.475*		*0.0*	mm^2/s
pv1	*0.19946*			bar(a)
pv2	*0.21085*			bar(a)
tv1	*311.74*	*99.974*	*99.974*	°C
cF1	*1,569.4*			m/s

Pipe downstream of valve

Size class downstream of valve	NPS2	*2"*
Schedule downstream	SCH2	*STD*

Valve configuration

Valve type		*Straight globe valve*
Trim type		*Contoured plug*
Flow direction		*FTC*
Valve performance class		*Multi stage valve*
Protection		*Non-hardened*
Low-noise design		*n.a.*

Valve data

Basic characteristic		*Linear*	
☐ Lift stopper			
Selected valve size	NPS	*2"*	
Pressure class	class	*class 600*	
Maximum operating pressure	p,max	*105.0*	bar(a)
Valve modifier (Cv/Cv100 = 0.75)	FL^2	*0.894*	-
Valve modifier (Cv/Cv100 = 0.75)	Kc	*0.846*	-
Cavitation modifier (Cv/Cv100 = 0.75)	xFz	*0.527*	-
Valve style modifier (Cv/Cv100 = 1)	Fd	*0.41*	-

Load-dependent values

	Maximum flow	Mean flow	Minimum flow	
u2	! *68.525*			m/s
LpAe	! *109.0*			dB(A)
Ri	! *7.5*			-

Noise calculation

Calculation standard (noise, liquid)	*IEC 60534-8-4 (2015)*

Noise prediction data

Low-noise design		*n.a.*	
Sound velocity	cF1	*1,569.4*	m/s
Pressure level	LA1	*184.0*	dB
Pressure level	LA2	*8.31*	dB
Sound pressure level	Lpi	*193.0*	dB

Noise prediction data (continued)

Transmission loss	TL	*-78.5*	dB
Sound power level (A-weighted)	LwAe	*120.0*	dB(A)
Sound pressure level of valve (A-weighted)	LpAe	*109.0*	dB(A)

Comments:

Reliability index - Ri

Sound pressure level: 70 %; High velocity: 29 %
Recommended solution: Heavy duty valve, Hardened seat/plug, Low-noise

Legend

Errors
Alarm